Subscription Churn: Business Analytics Framing Document

1. Business Objective

   Reduce short-term and medium-term subscriber churn by identifying early warning signals, predicting churn risk 7–14 days in advance, and deploying targeted retention actions that maximize retained users and revenue while minimizing unnecessary discounts.

2. Monthly Retention Rate

   (Primary success metric)

   Definition:
   Percentage of users who had an active subscription at the start of the month and are still active at the end of the month.

   Why this metric?

   - Directly reflects customer stickiness
   - Aligns across Product, Growth, and Revenue teams
   - Sensitive to both cancellations and failed renewals.

   Secondary proxy (for leadership):
   Net Revenue Retention (NRR – proxy)
   Since expansion data may be limited, we track **revenue retained vs revenue lost from churn**.

3. Supporting KPIs (Operational & Diagnostic)

   Core Churn & Retention Metrics

1. *Churn Rate (Weekly / Monthly)*
   % of users who churn in a given period
2. *Retention Rate / Cohort Retention*
   % of users retained by signup cohort over time (week 1, week 4, month 2, etc.)
3. *Renewal Success Rate*
   % of expiring subscriptions that successfully renew.

   Usage & Engagement Metrics

4. Average Usage Days per Week
   Mean number of active days per user per week
5. Average Usage Minutes per Week
   Measures depth of engagement (time spent).

Payments & Revenue Metrics

6. *Payment Failure Rate*
   % of payment attempts that fail (card declined, insufficient balance, expiry)

7. *Revenue Retained vs Revenue Lost (Proxy)*
   - Revenue retained from renewed users
   - Revenue lost due to churned subscriptions.

<u>Marketing / Retention Effectiveness</u>

8. *Campaign Response Rate*
   % of users who engage (click / redeem / renew) after a retention touchpoint

9. *Post-Campaign Renewal Lift (optional but powerful)*
   Renewal rate of touched users vs non-touched users

4. Explicit Churn Definition

A **user is considered churned if**:

> **The subscription is cancelled OR the subscription expires and is not renewed within 7 days after expiry.**

Notes:

- The 7-day grace window accounts for delayed payments and retry logic
- Churn is attributed to the **first day after the grace period ends**
- This definition will be used consistently across:
  - Analysis
  - Prediction model
  - Dashboards
  - Reporting

5. Scope & Time Windows

Window (Feature Lookback)

Past 30 days of user data used to generate features:

- Usage behaviour (minutes, sessions, active days)
- Payment attempts & failures
- Campaign touchpoints
- Subscription status & tenure

Rationale:
30 days balances behavioural signal strength with freshness.

Prediction Window

Next 14 days

Target variable:

"Will this user churn in the next 14 days?"

Used for:

- Early-warning alerts
- Retention campaign triggers
- Risk-based user segmentation

6. Stakeholder Questions This Analysis Will Answer

1. *What are the primary drivers of churn?*
   (Usage drop, pricing, plan type, payment failures, tenure)
2. *How early can we reliably identify users at risk of churning?*
   (7–14 days in advance using behavioural and payment signals)
3. *Which users should we prioritize for retention to maximize ROI?*
   (High-risk vs medium-risk vs low-risk users)
4. *What role do payment failures play in churn, and how much churn is preventable?*
5. *Which subscription plans and price points are most vulnerable to churn?*
6. *Do retention campaigns actually reduce churn, and for which segments?*
7. *What is the expected reduction in churn and revenue loss over the next 30 days if we act?*

8. Success Criteria (How Leadership Will Judge This)

- Clear identification of **top churn drivers**
- Actionable churn prediction with strong recall on churners
- Measurable improvement in:
  - Retention rate
  - Renewal success
  - Revenue retained
- A retention strategy that is **targeted, not blanket discounting**